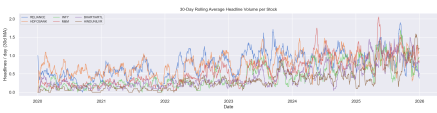
Multi-Dimensional Return Forecasting and Portfolio Management

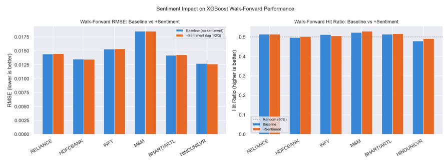
Data, Feature Engineering & Sentiment Pipeline

- **Dataset Composition:** Pipeline utilizes approximately 1,486 trading days worth of data per stock (Jan 2020 to Dec 2025) across 6 sector-diverse NSE large-caps (BHARTIARTL, HDFCBANK, HINDUNILVR, INFY, M&M, RELIANCE).
- **Alternative Data Pipeline (Sentiment):** Financial news fetched via Google News RSS and scored using `ProsusAI/finbert` Headlines are classified into positive, negative, and neutral confidence scores.
- **Sentiment Aggregation:** Daily aggregations include `sentiment_mean`, `pos_count`, `neg_count`, `neutral_count`, and `headline_volume`. Gaps forward-filled (≤ 5 days); remaining NaNs fall back to neutral (0).
- **Technical Space (40+ Features):** Short-lag returns (1–10d), rolling volatility/mean (5–20d), SMA/EMA, RSI(14), MACD, Bollinger Bands, volume z-scores, and lagged fundamental data.

Sentiment Time Series



Headline Volume

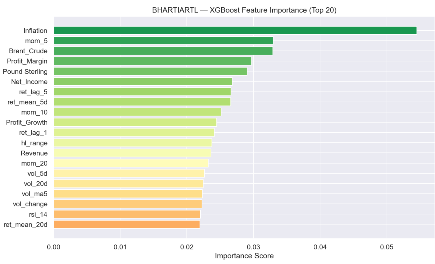


Sentiment Comparison

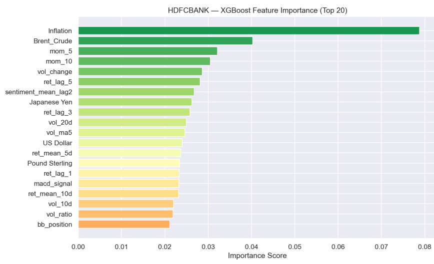
Figure 1: Alternative data integration showcasing the sentiment signals driving the models.

Model Architecture, Validation & Feature Importance

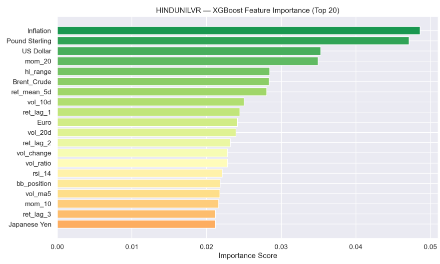
- **Model Ensemble:** Evaluated XGBoost, LightGBM, and a 2-layer LSTM model. Boosted trees outperformed LSTM.
- **Walk-Forward Validation:** 5-fold expanding-window cross-validation strategy (Train: Jan 2020–Sep 2025; Test: Oct–Dec 2025).
- **Target Formulation:** $Target = \ln(Close_{t+1}/Close_t)$ (Next-day log returns).



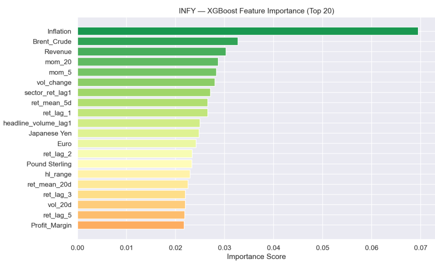
BHARTIARTL



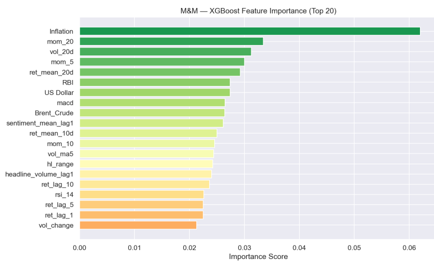
HDFCBANK



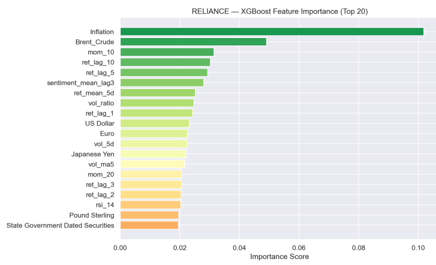
HINDUNILVR



INFY



M&M

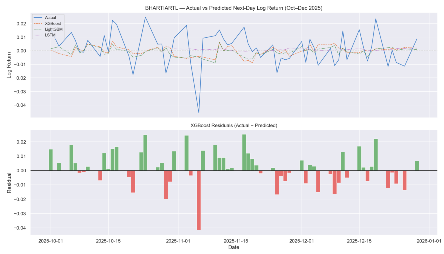


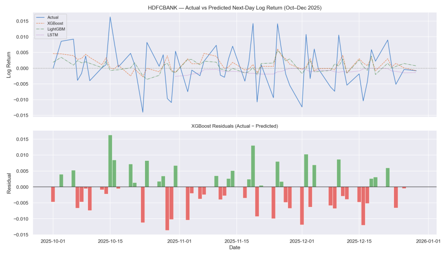
RELIANCE

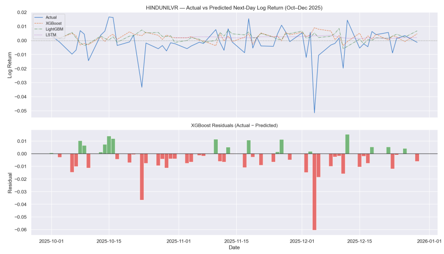
Figure 2: Top influential features for all stocks, emphasizing volatility and sentiment dynamics.

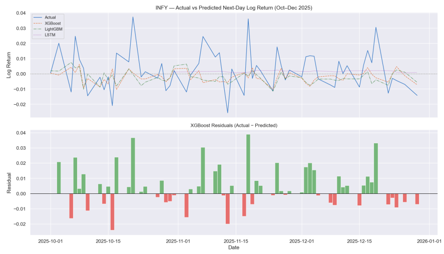
Table 1: Champion models and out-of-sample directional hit performance.

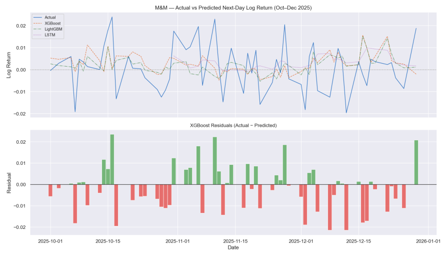
Stock	Best Model	WF Hit %	Test Hit %
BHARTIARTL	XGBoost	52.4%	60.0%
HDFCBANK	XGBoost	50.7%	40.0%
HINDUNILVR	LightGBM	50.7%	47.5%
INFY	LightGBM	51.2%	45.0%
M&M	LightGBM	52.7%	62.5%
RELIANCE	XGBoost	51.9%	50.0%


BHARTIARTL


HDFCBANK


HINDUNILVR


INFY


M&M

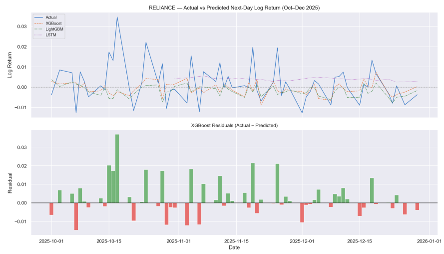
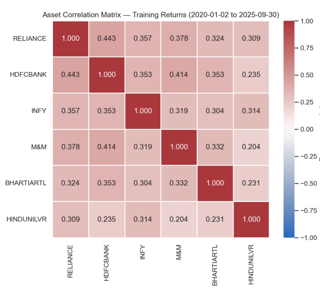

RELIANCE

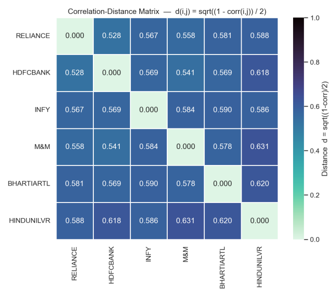
Figure 3: Actual vs. Predicted return scatter plots capturing OOS variance.

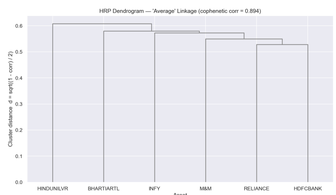
Portfolio Construction & Evaluation (Oct–Dec 2025)

Three allocation schemas evaluated on identical data (Test window: 2025-10-01 to 2025-12-31):

- **Equal-Weight (EW):** Naive static baseline ($1/N$).
- **Risk-Adjusted (RA):** $w_i \propto \hat{r}_i/\hat{\sigma}_i$, long-only, **rebalanced daily**.
- **Hierarchical Risk Parity (HRP):** Forecast-free allocation driven entirely by historical correlation and variance.


Correlation Heatmap


Distance Matrix


HRP Dendrogram

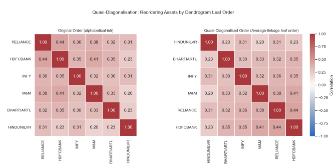
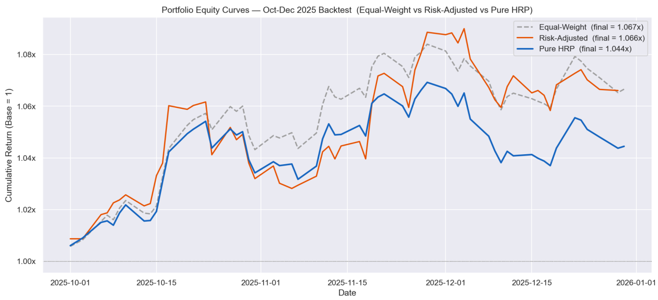

Quasi-Diagonalized Cov.

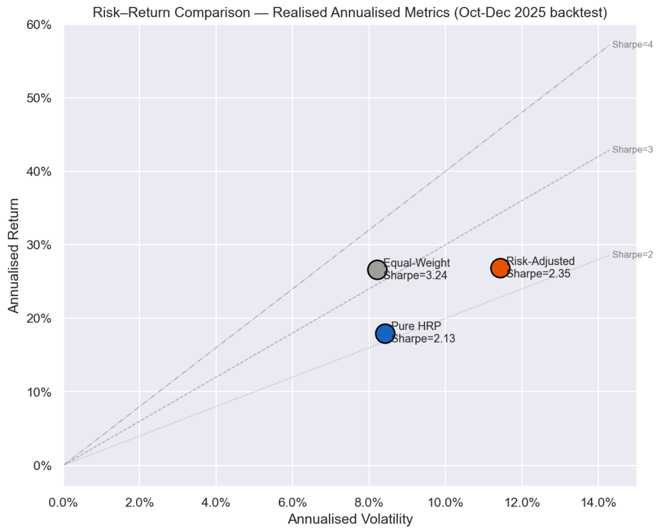
Figure 4: HRP Clustering mechanism used to derive stable correlation-based portfolio weights.

Table 2: Portfolio performance backtest (Out-of-Sample: Oct–Dec 2025).

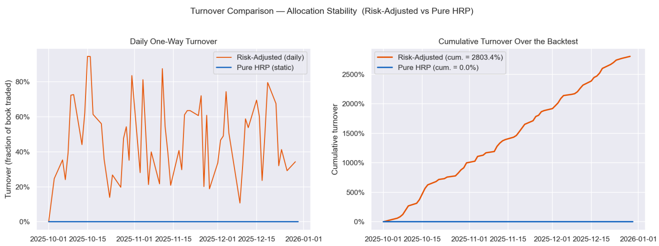
Metric	Equal-Weight	Risk-Adjusted	Pure HRP
CAGR (Annualized)	30.51%	30.83%	19.69%
Annual Volatility	8.22%	11.43%	8.42%
Sharpe Ratio	3.24	2.35	2.13
Sortino Ratio	7.44	4.09	4.29
Max Drawdown	-2.34%	-3.15%	-3.01%
Cumulative Turnover	0.00%	2803.41%	0.00%



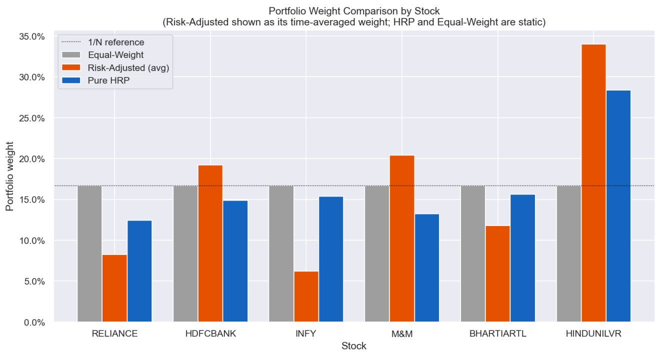
OOS Equity Curves



Risk vs Return Comparison



Cumulative Turnover



Portfolio Weights Comparison

Figure 5: Portfolio Aggregation results capturing equity growth, risk characteristics, turnover limits, and asset allocations.

Key Quality Controls

- **Leakage Prevention Audit:** Causal shift-aligned rolling windows; per-fold scaling metrics; HRP computed strictly from pre-test history.
- **Reproducibility:** Global seeds fixed, ensuring robust and auditable artefacts.